

RAFFLES INSTITUTION
2021 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
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BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

15th Sept 2021
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question on the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, **hand in your essay question SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 32
2	/ 8
3	/ 10
Section B	
4 or 5	/ 25
Total	/ 75

This document consists of **15** printed pages.



Raffles Institution
Internal Examination

Section A

Answer **all** the questions in this section.

- 1** The COVID-19 pandemic is caused by the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), which was first identified in December 2019. As of June 2021, more than 180 million cases of infection and more than 3.9 million deaths have been confirmed worldwide.

The severity of COVID-19 symptoms is highly variable, ranging from asymptomatic to life-threatening. COVID-19 is transmitted via droplets in the air, small airborne particles and via contact with contaminated fluids.

Fig. 1.1 show the structure of the SARS-CoV-2 virus, which contains a single-stranded RNA genome.

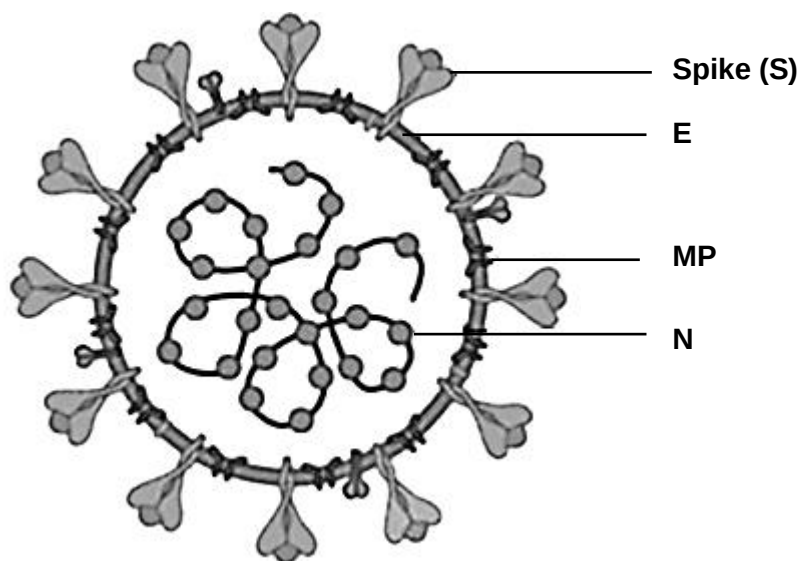
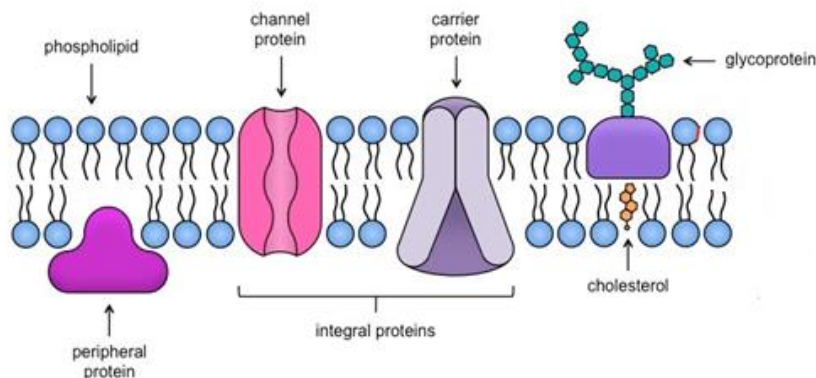


Fig.1.1

- (a)** Identify structure **E** and draw a labelled diagram showing its structure in the space below. [3]

1. Structure **E**: Envelope/phospholipid bilayer (must have);



Labeled diagram showing:

2. Bilayer (compulsory pt); R: double bilayer
3. Phospholipid (ball + 2 sticks);
4. Glycoproteins / glycolipids / spike protein / cholesterol;

Fig. 1.2 shows the replication cycle of the SARS-CoV-2 virus.

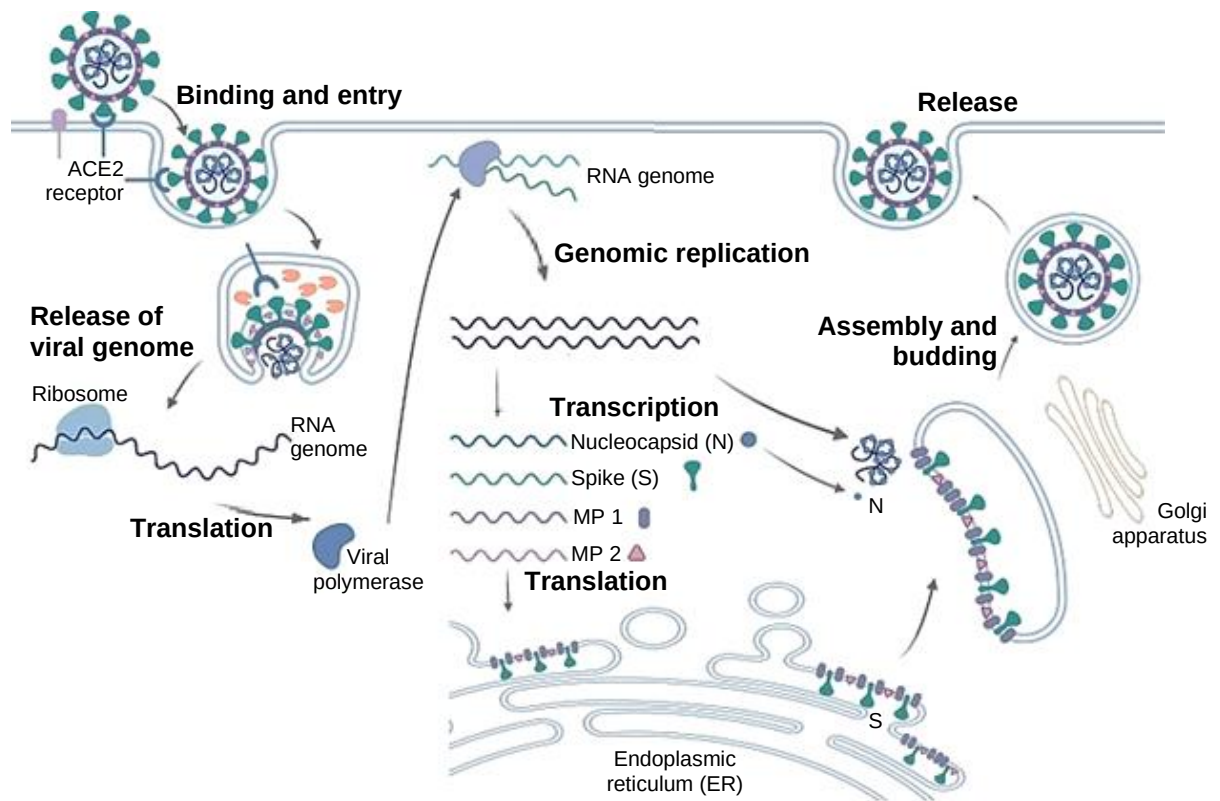


Fig. 1.2

- (b) Describe how genome replication differs between the SARS-CoV-2 virus and the T4 bacteriophage. [2]

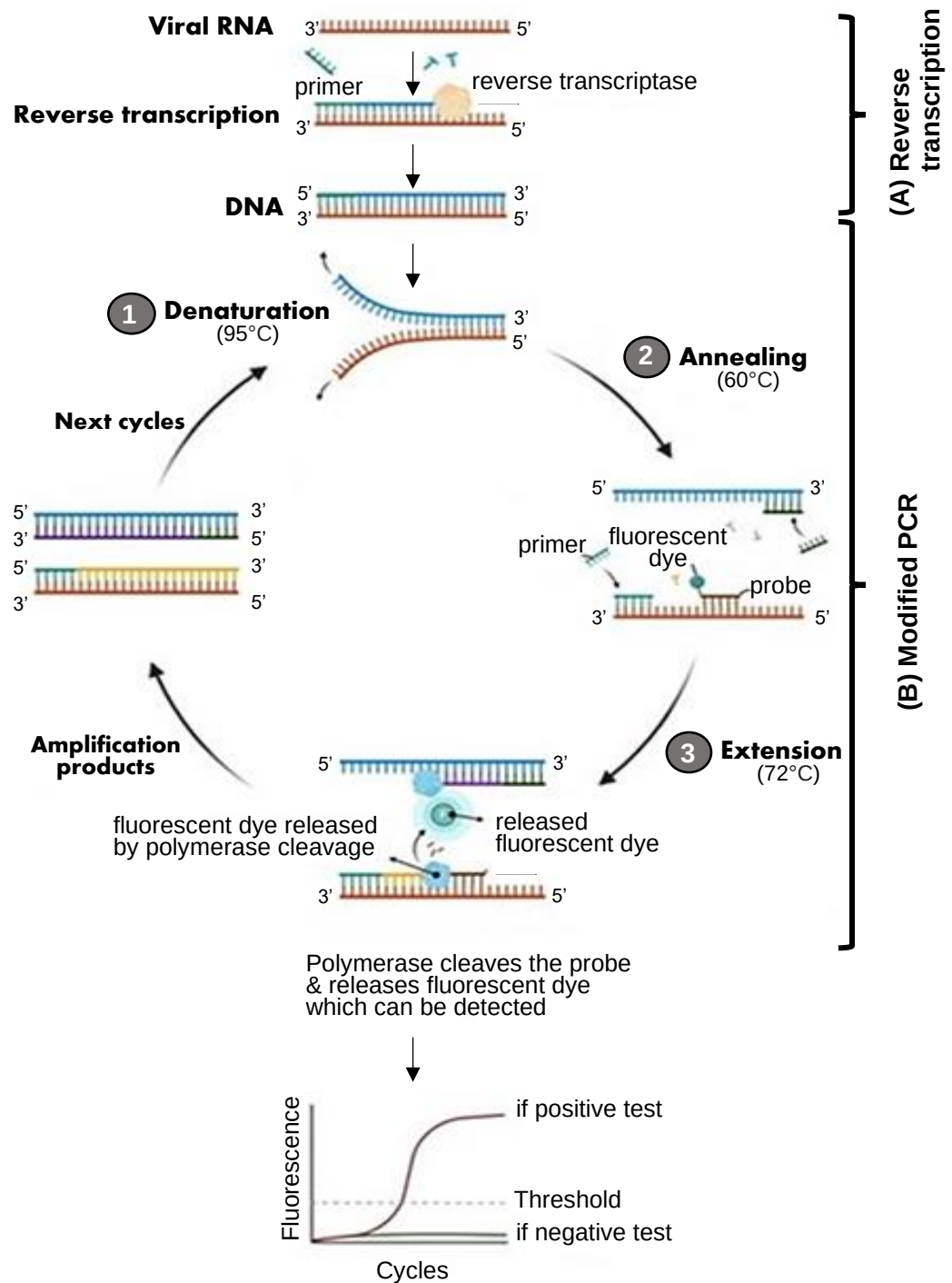
Event	SARS-CoV-2	T4
1. Template for genome replication R: if context is about transcription/translation	Uses <u>single-stranded RNA as a template*</u>	Uses <u>double-stranded DNA as a template*</u>
2. Product	To <u>synthesize single-stranded RNA</u>	To <u>synthesize double-stranded DNA</u>
3. Substrate used	Use ribonucleotides to make RNA	Use deoxyribonucleotides to make DNA
4. Enzyme involved R: if context is wrong e.g. use viral RNA polymerase to synthesis DNA	Uses <u>viral RNA dependent RNA polymerase</u> ; Accept <u>viral polymerase</u> as it is shown in the diagram R: RNA polymerase because it can refer to host RNA polymerase	Uses <u>host DNA polymerase</u>
5. AVP	Needs to synthesise viral RNA polymerase first before genome replication OWTTE	Use host polymerase directly for genome replication OWTTE
If written as one para vs another para. Only 1 mark and only if entire para is correct.		

The COVID-19 test is used to diagnose active infection with the SARS-CoV-2 virus.

The test involves the following steps:

- Insertion of a cotton swab through the patient's nostrils into the nasopharyngeal space so that a swab sample is collected.
- Extraction of viral genetic material using an extraction buffer.
- Reverse transcription and Polymerase Chain Reaction.

Fig. 1.3 below shows the process of (A) reverse transcription, followed by (B) modified PCR used in the COVID-19 test.



- (c) (i) Explain why the (A) reverse transcription process is required before carrying out the (B) modified PCR. [2]
[what is reverse transcription]
1. Reverse transcription used RNA as a template to synthesise DNA;
[why there is a need to make DNA from the RNA template]
 2. DNA is used as a template for modified PCR as Taq polymerase is DNA dependant DNA polymerase;
 3. AVP
- (ii) With reference to the (B) modified PCR process shown in Fig. 1.3, explain the steps labelled: denaturation; [2]
1. Heating to 95°C causes the weak hydrogen bonds* between complementary bases of each strand to break (R: denature) due to increased molecular vibrations;
 2. The denaturation of double-stranded DNA into 2 single-stranded DNA;
 3. To expose the sequence for primers to anneal;;
Or
So that each strand can act as a template* for synthesis of the complementary strand
- primer annealing. [2]
1. Lower temperature of 60°C allows primers to bind/ anneal specifically via complementary base pairing*;
 2. to the regions flanking the target DNA sequence and thus determine the segment to be amplified;
 3. and provides a free 3'-OH group for chain extension;
- (iii) Compare the primers and probes used in the (B) modified PCR process shown in Fig. 1.3. [2]

At least 1 similarity and 1 difference

Event	Primers	Probe
Number	1. Two types (forward and reverse primers)	One type
Sequence	2. <u>Different sequence</u> or <u>Complementary to both strand</u>	<u>One single sequence</u> or <u>Complementary to one strand only</u>
Fluorescent dye	3. <u>No fluorescent dye attached</u>	<u>Fluorescent dye attached</u>
Location where it binds	4. <u>Bind to the 3' end of template strand</u>	<u>Bind to the middle of the template strand</u> R: anywhere on the template strand; target sequence, complementary sequence
Composition	5. Both made of <u>deoxyribonucleotides/DNA</u> joined by <u>phosphodiester bond</u> (A: without phosphodiester bonds) or <u>Both short ssDNA/oligo nucleotides</u> ;	
Complementary to	6. Both are complementary to /anneal to DNA (A: viral DNA)	

Source	7. Both are <u>chemically/artificially synthesized</u>	
AVP- difference	8. The function is to provide 3' OH group for chain extension to amplify the target sequence	The function is to release fluorescent dye to indicate the presence of target sequence
AVP - similarity		

Scientists across the globe have worked rapidly to develop an effective vaccine against SARS-CoV-2. The SARS-CoV-2 spike protein has been identified as an effective target for use in vaccine design. Vaccines using the spike protein can be prepared via a variety of approaches.

Fig. 1.4 shows two different approaches.

- In the replication-defective viral vector vaccine, the DNA containing the gene that codes for the SARS-CoV-2 viral spike protein is enclosed in a viral vector capsid.
- In the lipid nanoparticle mRNA vaccine, the mature mRNA transcribed from the gene coding for SARS-CoV-2 viral spike protein is enclosed in a lipid nanoparticle coat.

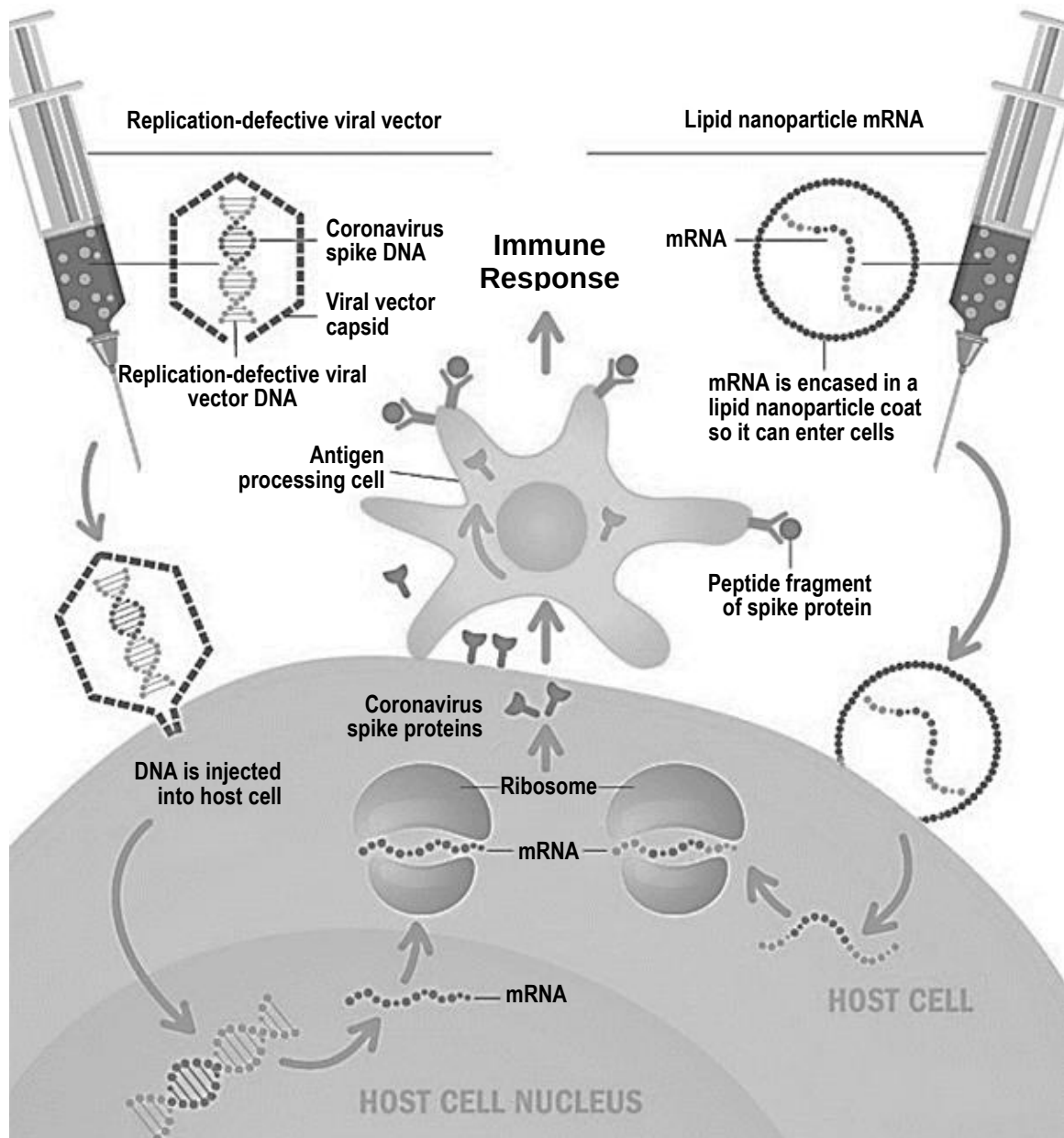
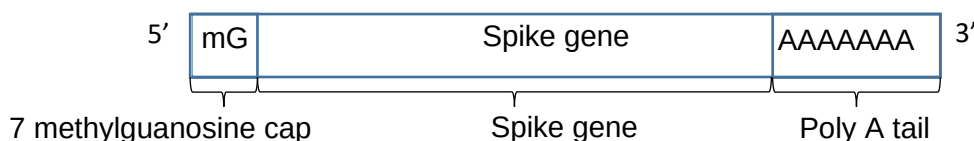


Fig. 1.4

- (d) (i) With reference to the lipid nanoparticle mRNA vaccine shown in Fig. 1.4, explain why mRNA is unable to enter the cell on its own but can do so when encased within a lipid nanoparticle coat. [2]

1. mRNA is too large, unable to pass through the transient pores of cell surface membrane;
 2. mRNA is hydrophilic/charged, unable to pass through hydrophobic core of phospholipid bilayer of the cell surface membrane;
- (Max 1)
3. Lipid nanoparticle coat, fusion* with phospholipid bilayer/ cell surface membrane, deliver mRNA into cytosol/cytoplasm;

- (ii) Using your knowledge of mature mRNA, draw a labeled diagram to show the structure of the mature mRNA that was delivered into the cell. [2]



1. Single line or block showing that mRNA is single stranded + polarity;
 2. 3' : Poly A tail;
 3. 5' : methyl guanosine cap;
- (iii) Explain how the spike proteins are produced after the translation initiation complex is formed. [4]
1. This positions initiator tRNA at peptidyl-tRNA binding site (P site)* leaving aminoacyl-tRNA binding site (A site)* vacant for incoming aminoacyl-tRNA molecules;
 2. A second aminoacyl-tRNA with a specific anticodon* and corresponding amino acid, binds to specific mRNA codon at A site via complementary base-pairing*;
 3. A peptide bond* is formed between adjacent amino acids, catalysed by peptidyl transferase* of large ribosomal subunit. Methionine dissociates from initiator tRNA as a result and remains bound to second amino acid at A site;
 4. The ribosome translocates in 5' to 3'* direction, shifting first tRNA to exit site (E site) allowing it to be released into cytosol. tRNA with growing polypeptide chain is now at P site and A site will hold a new incoming aminoacyl-tRNA with an anticodon complementary to next codon on mRNA;
 5. This process continues until a stop codon (UAA/ UAG/ UGA)* is reached at A site. Release factors* will enter A site
 6. causing hydrolysis of bond between polypeptide chain and tRNA in P site; Polypeptide is released from ribosome;
 7. Include RER folding /Golgi Apparatus markpoint;

- (iv) Suggest one advantage of using mRNA in the vaccine. [1]
1. Unlike DNA vaccines, mRNA vaccine is short-lived/degraded by RNase over time, no long time effect;
 2. Unlike the replication-defective viral vector DNA vaccine, mRNA does not get integrated into the host cell genome/chromosome, no long time effect;
 3. Unlike the replication defective viral vector DNA vaccine, mRNA vaccine does not contain a virus vector, no chance of resistance/immune response mounted against the mRNA vaccine;
 4. No chance of reversion of virulence;
 5. AVP
- (v) Explain how the expression of spike proteins in a vaccinated individual protects against SARS-CoV-2 infection. [5]
- Immune response (remove) (max 4)**
1. Viral spike proteins stimulate an immune response;
 2. Antigen presenting cells (APCs) macrophages / dendritic cells (A: antigen processing cells) take up the spike protein by phagocytosis, process antigen and present it as a peptide:MHC complex;
 3. Each naïve T cell has a specific T cell receptor (TCR) that specifically binds to complementary peptide:MHC complex on APC;
 4. APC secretes cytokines* that activates naïve T cell which will undergo clonal expansion and differentiation to form helper T cells, cytotoxic T cells and memory T cells;
 5. T helper cells* bind to secrete cytokines that activate specific naïve B cells* to undergo clonal expansion and differentiation and form antibody-secreting plasma cells and memory B cells;
- How you are protected (mechanism) (max 2)**
6. Memory B and T cells* when exposed to virus, will recognize it and mount a faster and stronger and longer lasting secondary immune response;
 7. Specific antigen binding site of antibody binds to specific epitope on spike protein of incoming virus;
 8. Prevent/block spike protein from binding to ACE receptors, resulting in neutralization*;
 9. Prevents attachment and subsequent infection of cells;
- Fc region/constant region of heavy chains bind to Fc receptors on phagocytes to tag / promote phagocytosis which is opsonisation*;

- (e) From the time of the first discovery of the SARS-CoV-2, different variants have emerged. Among them, the Delta variant, also known as lineage B.1.617.2, is of international concern as it has been shown to have a higher transmission rate.

Sequence analysis revealed significant changes to the spike proteins. Table 1.1 shows these changes.

Table 1.1

position of amino acid	amino acid change	
	from	to
478	threonine	lysine
452	leucine	arginine
681	proline	arginine

- (i) Suggest and explain why these changes in amino acids have arisen. [3]
1. Single-stranded RNA genome;
 2. No backup copy to do correction;
- OR
3. (RNA-dependent/ viral) RNA polymerase lacks proof-reading ability;
 4. error in viral genome replication;
- OR
5. RNA genome more reactive (due to the 2'OH group); R: unstable
 6. result of mutations of the RNA* genome leading to the changes in the RNA sequence (known as antigenic drift);
 7. change in codon* sequences resulting in a change in type of amino acids;
- (ii) Suggest an explanation for the increase in the Delta variant transmission rate. [2]
1. Mutant spike proteins has shape that is more complementary in shape to ACE-2 receptor;
 2. Binds at higher affinity to the ACE-2 receptor;
 3. Increase efficiency of infection of host cells, resulting in greater transmission / lower viral load is required for infection;
 4. More virions will be released to infect others (cells/hosts);
 5. AVP e.g. the mutant spike proteins are less complementary in shape to the antigen-binding sites of the antibodies and able to evade the immune response;

[Total: 32]

2

(a)

Fig. 2.1 shows an artificially synthesized oligonucleotide binding to the promoter region of the DNA sequence, forming a triple helix (triplex).

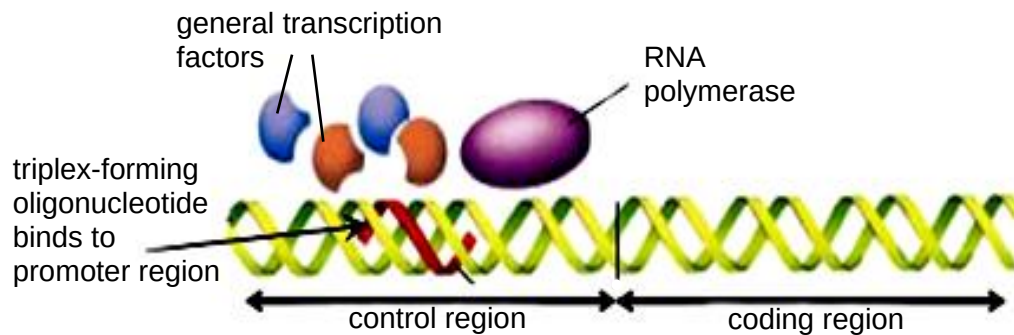


Fig. 2.1

(i) With reference to Fig. 2.1, explain why transcription cannot occur in the presence of the artificially synthesised oligonucleotide. [2]

1. The DNA binding site on the general transcription factors (GTFs)* are no longer complementary in shape to the promoter region because of the formation of the triplex;
Or
Oligonucleotide binds to the promoter and blocks binding of general transcription factors*;
2. GTFs unable to bind to the promoter and recruit RNA polymerase* to form the transcription initiation complex* for initiation of transcription;

Fig. 2.2 shows another artificially synthesized oligonucleotide binding to the coding region of the DNA.

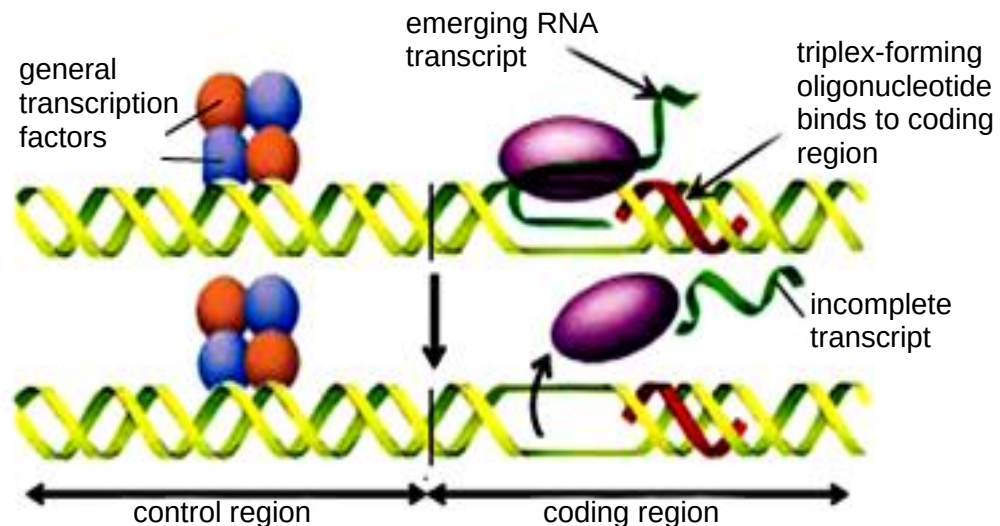


Fig. 2.2

(ii) Describe how gene silencing occurs in Fig. 2.2. [2]

EITHER

1. The oligonucleotide disrupts the binding of RNA polymerase as it moves down the template DNA strand;

OR

Oligonucleotide prevents RNA nucleotides from being added to the RNA chain via complementary base-pairing with DNA nucleotides;

OR

Oligonucleotide prevents RNA polymerase from separating the double-stranded DNA molecule to single-stranded so that one strand can be used as a template for transcription.

EITHER

2. causing transcription to be terminated early and the mRNA formed to be incomplete or truncated;

OR

Translation of truncated mRNA will produce non-functional protein or no protein produced.

Gene expression patterns can be analysed using polymerase chain reaction (PCR).

Methanolic extracts of *Imperata cylindrica* leaf (ICL) have been shown to affect the expression of the *CASP8* gene in human oral squamous carcinoma cells. *CASP8* gene codes for a caspase protein which plays a key role in activating programmed cell death. Human oral squamous carcinoma cells were treated with two different concentrations of methanolic extracts from ICL and the expression of the *CASP8* gene in these cells was measured.

The relative expression of the *CASP8* gene was measured from oral squamous carcinoma cells without treatment (control), with 320 µg/ml ICL treatment and with 640 µg/ml ICL treatment.

The relative expression of the *CASP8* gene can be seen in Fig. 2.3.

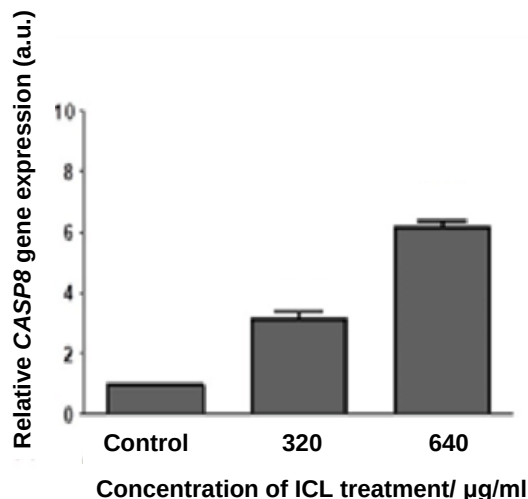


Fig. 2.3

- (b) (i) Using the information in Fig. 2.3, describe the results of ICL treatment on the human oral squamous carcinoma cells. [2]
1. Trend: Increase in the concentration of ICL treatment will result in an increase in the relative CASP8 gene expression;
 2. Fig. 2.3 shows that expression of CASP8 gene is 6 a.u. for cells treated with 640 µg/ml ICL compared to 3 a.u for cells treated with 320 µg/ml ICL; (A: 2x more);
 3. The expression of CASP8 gene is very low in the control cells as seen by the low levels of expression at 0.9/1 a.u.;
- (ii) Explain the effectiveness of ICL in the treatment of human oral squamous carcinoma. [2]
1. The ICL extract is able to increase expression of CASP8 gene in the human oral squamous carcinoma cells resulting in an increase in concentration of caspase proteins;
 2. which will cause the cancerous cells to undergo apoptosis/programmed cell death;
 3. This makes the ICL extract an effective anticancer drug;

[Total:8]

- 3 Mitochondrial DNA (mtDNA) is a tool for determining the origin of different populations. It is also an important object of study in different fields such as medical genetics, genetic genealogy and evolutionary phylogeny.

MELAS syndrome is an inherited disease in humans caused by a mutation in a gene located in the mtDNA.

Fig. 3.1 shows four generations of a family where several individuals are affected by MELAS syndrome.

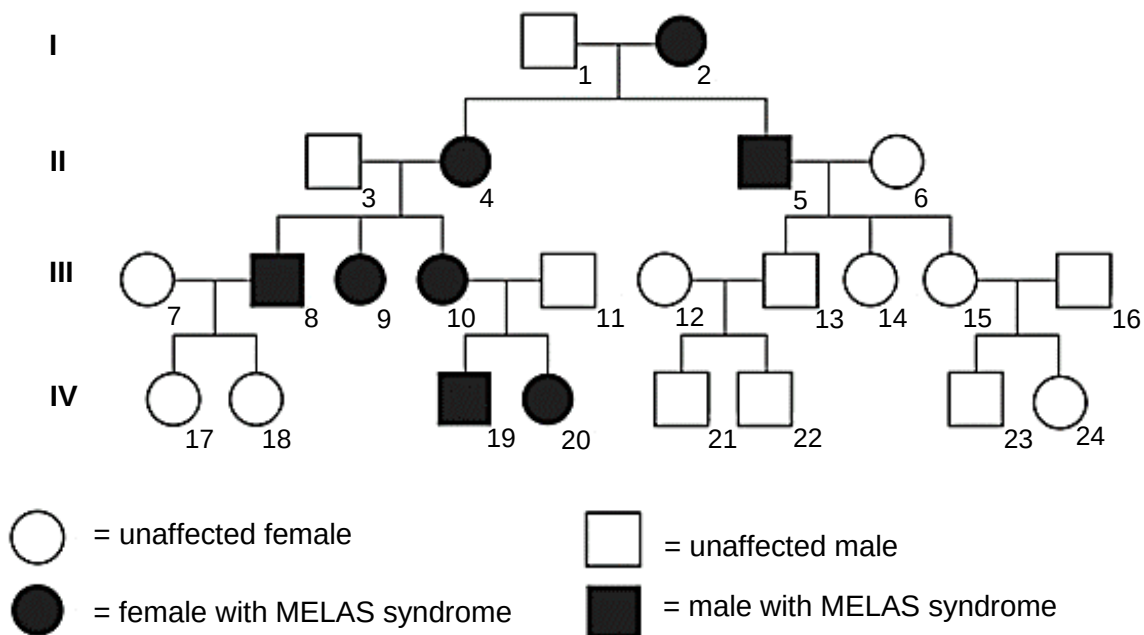


Fig. 3.1

- (a) (i) Explain, using evidence from Fig. 3.1, why MELAS syndrome is a mtDNA disease. [3]
1. mt DNA is maternally inherited as the mother's egg contributes the cytoplasm including the mitochondria while the father's sperm contributes only the nucleus to the zygote;
 2. Inheritance is through mtDNA as all offspring of a female with MELAS syndrome also have MELAS syndrome;
 3. But none of the offspring of affected males with MELAS (with unaffected wives) have MELAS syndrome;
 4. ref. to numbered individuals to support;
 mp1: 17 and 18 are unaffected although their father, 8, is affected
 13, 14 and 15 are unaffected although their father, 5, is affected
 OR
 mp2 : 4 and 5 are affected as their mother, 2, is affected
 8, 9 and 10 are affected as their mother, 4, is affected
 19 and 20 are affected as their mother, 10, is affected

Must mention mt DNA is maternally inherited to get points 2, 3 and 4

- (ii) Describe and explain characteristics of mtDNA that makes it more useful than using nuclear DNA in determining evolutionary relationships. [2]
1. mtDNA is a single copy of DNA /haploid it does not undergo crossing over/recombination;
 2. hence any differences in mt DNA is due to mutations;
 3. mtDNA undergo a higher rate of mutations which occur at clockwork regularity;
 4. hence mtDNA can distinguish between closely related organisms;
 5. hence mtDNA can more accurately estimate timing of speciation:

Pts: 1+2...OR Pts: 3 +4/5

- (b) Giant pandas, *Ailuropoda melanoleuca* and red pandas, *Ailurus fulgens* are both mammals which are native to China. It was long believed that giant pandas and red pandas were close evolutionary relatives and there is much evidence to support this hypothesis.

- Both animals live in similar habitats.
- Both have digestive systems similarly adapted to their bamboo diet.
- Both have a sixth digit, known as a pseudo-thumb, which they use to grip and shred bamboo shoots.



Fig. 3.2

With the advent of DNA sequencing techniques, it has been possible to compare the DNA of different species to confirm how closely related they are to each other. The table below shows mtDNA sequences from four species of mammals including that of the giant panda and the red panda.

Table 3.1

species	mtDNA codes
black bear	ATTGGAGCAGACTTA.....
giant panda	ATTGGCACTAATCTA.....
red panda	ATTGGAACTAATCTT.....
raccoon	ATCGGAACTAATCTT.....

- (i) Use Table 3.1 to identify which of the species is most closely related to the red panda.

Explain your answer. [2]

1. Raccoon;
2. there is only a 1 base difference between the raccoon and the red panda compared to the red panda and the giant panda which have a 2 base difference or red panda and black bear which have a 6 base difference.

In the study, mtDNA base pairs were analysed.

The following number of differences were found between the DNA of the giant panda and that of the other species analysed:

red panda 17
black bear 12
raccoon 21

An estimate of the mutation rate for the mtDNA sequence analysed is 3.95×10^{-7} mutations per year.

- (ii) Calculate how many years ago the giant panda and the red panda last shared a common ancestor. Give your answer in **standard form**, to **two significant figures**. [2]

3.95×10^{-7} mutations $\rightarrow$ 1 year

17 differences/mutations $\rightarrow 1 / 3.95 \times 10^{-7} \times 17$;

= 43037974.68

= 43 000 000

= 4.3×10^7 years;(2 sig. fig)

Max 1 mark if answer not given in standard form and 2 sig fig.

- (c) In addition to mtDNA sequences, information based on the presence of shared characteristics in various organisms can also be used to show relatedness and to draw phylogenetic trees.

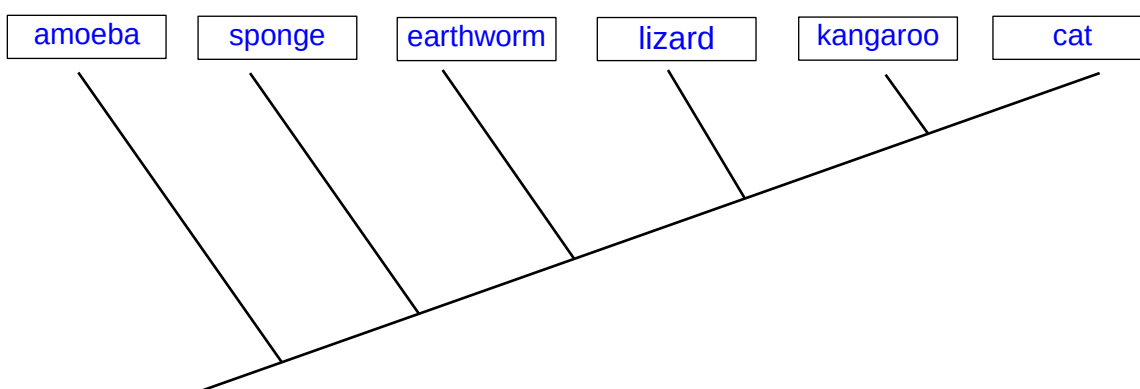
Table 3.2 shows certain characteristics present (+) or absent (-) in 6 organisms.

Table 3.2

	segmented	jaws	hair	placenta	multicellular	limbs
amoeba	-	-	-	-	-	-
cat	+	+	+	+	+	+
earthworm	+	-	-	-	+	-
kangaroo	+	+	+	-	+	+
lizard	+	+	-	-	+	+
sponge	-	-	-	-	+	-

Using the data, label the cladogram with the names of the organisms.

[1]



All correct [1]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a) Apart from enzymes, describe the significance of various types of proteins in the regulation of processes and responses in cell. [15]

1. Regulation of gene expression

A. Histones

1. Addition of **acetyl groups*** to lysine residues on **histones*** removes positive charge from histones thus reducing electrostatic interaction between negatively charged DNA and histone and DNA winds less tightly around histones;
2. **This allows **RNA Polymerase*** and general **transcription factors*** to bind the **promoter*** to form the **transcription initiation complex***.
3. Removal of **acetyl groups*** to lysine residues on **histones*** restores positive charge on histones thus increasing electrostatic interaction between negatively charged DNA and histone and DNA winds more tightly around histones;
4. **This prevents **RNA Polymerase*** and general **transcription factors*** from binding to the **promoter*** to form the **transcription initiation complex***. [significance].
5. **Methylation of histones** occurs on coding regions containing genes coding for proteins that are not required in the cell to tightly package DNA into heterochromatin, preventing expression of such genes [significance];

B. General transcription factors

1. **Binding of **general transcription factors*** and **RNA polymerase*** to promoter region to form the **transcription initiation complex*** / initiate **transcription*** [significance].

C. Specific transcription factors

1. **Activators*** bind to **enhancers***, promoting the assembly of **transcription initiation complex*** / increasing the frequency of transcription. [significance].
2. Description of any of the mechanisms by which activators upregulate gene expression/increase frequency of transcription:
 - a. Activator interaction with enhancer results in the bending of spacer DNA which allows direct interaction of **activators*** with RNA polymerase and/or general transcription factors at the promoter,
 - b. Activator may recruit **histone acetyltransferase*** / **chromatin remodeling complex** to decondense chromatin.
3. Description of any of the mechanisms by which repressors downregulate gene expression/decrease frequency of transcription:
 - a. Enhancer region overlaps with silencer region. Therefore, binding of repressor prevents binding of activator.

- b. Repressor binds to activator to prevent it from interacting with general transcription factors.
- c. Repressor interacts with general transcription factors to prevent assembly of transcription initiation complex.
- d. Repressor may recruit histone deacetylase* / chromatin remodeling complex to condense chromatin.

- 4. Repressors* bind to silencers* which prevent the assembly of the transcription initiation complex* / decreasing the frequency of transcription [significance].

E. Prokaryotic repressors and activators

- 1. Active lac/trp repressor* binds to operator*, denying access of RNA polymerase to the promoter, preventing the transcription of structural genes. [significance].
- 2. Active catabolite activator protein* binds to the CAP-binding site of promoter*, increasing the affinity of the promoter for RNA polymerase*, increasing transcription frequency of the structural genes.
- 3. Tryptophan* acts as a corepressor to activate trp repressor*.

F. Eukaryotic initiation factors

- 1. Eukaryotic initiation factors* recognise and bind to 5' cap and/or 3' poly-A tail.
- 2. Eukaryotic initiation factors* recognise and bind to small ribosomal subunit and aminoacyl-tRNA.
- 3. Together with the large ribosomal subunit, allows for the formation of the translation initiation complex* / translation initiation [significance].

G. Ubiquitin

- 1. Protein molecules that tag the protein to be degraded for it to be sequestered into the proteasome* for degradation by hydrolytic enzymes [significance].

I. Release Factors

- 1. Release factors enter the A site resulting in the hydrolysis of the bond between the polypeptide chain and the tRNA in the P site.
- 2. The release of the polypeptide chain allows it to fold into a protein.

**Points 1A2, 1A4, 1B1 and 1C2 are similar in concept and only one mark was awarded even if more than one of them was written.

2. Cell signaling

A. Ligands

- 1. Protein ligands are complementary in shape and charge* to a specific ligand-binding site* of protein receptor to form a ligand-receptor complex.
- 2. causes a conformational change in the intracellular domain of the protein receptor which initiates signal transduction [significance].
- 3. E.g. insulin, glucagon and cellular response

R: non-protein ligands

B. Receptor

1. Receptor proteins bind and transduce the message of the signal molecule into a cellular response. [significance].
2. The intracellular domain/cytoplasmic side of GPCR has a **G protein binding site*** that allows binding of a G protein complex and activation of downstream relay molecules [significance].
3. E.g. glucagon receptor and how it receives the signal from the ligand and activates downstream relay molecules.
4. Phosphorylated **tyrosine*** residues on intracellular domain of RTK is recognised by specific relay proteins which bind and in the process are themselves activated [significance].

R: G protein due to GTPase function.

3. Cell cycle

A. Proto-oncogene protein products

1. Proto-oncogene protein products stimulate normal cell growth and division [significance].
2. e.g. growth factors, growth factor receptors, transcription factors, ras (relay protein), etc.

B. Tumour suppressor proteins

1. Tumour suppressor proteins which inhibit cell division and helps to prevent **uncontrolled cell division*** [significance].
2. Tumour suppressor proteins activate cell cycle arrest, DNA repair and/or apoptosis (programmed cell death).
3. e.g. p53 – activator which promotes synthesis of cell cycle-inhibiting proteins

AVP: Ref to cyclins binding to Cdks to activate Cdks → concentrations control the progression of the cell cycle [significance].

4. Transport mechanisms

A. Channel proteins

1. Transmembrane **channel proteins**^Δ with **hydrophilic*** pores that allow charged/polar molecules to diffuse across the membrane. * [significance].
2. e.g. aquaporins, Na⁺ channels (named example + molecule that diffuses through it), ATP synthase

B. Carrier proteins

1. Carrier proteins are transmembrane **carrier proteins*** that undergo conformational changes to allow molecules to be transported across the membrane. [significance].
2. e.g. Na⁺/K⁺ pumps, GLUT transporters (named example + molecule that diffuses through it)

5. Immunology

A. Cytokines

1. **Cytokines*** and chemokines result in increased permeability of blood vessels, to recruit neutrophils from blood to the site of infection. [significance].
2. Cytokines are signalling molecules released to stimulate proliferation and differentiation [significance].

*To get this point, you needed to have clearly stated that cytokines are proteins.

B. Immunoglobulins

1. Immunoglobulins contain antigen bindings sites* that are complementary in shape* to the specific antigen.
2. Immunoglobulins contain Fc region* that are complementary in shape and charge to Fc receptors* on phagocytes.
3. Allows for:
 - a. Neutralisation* where antigen binds to the antigen binding site of the antibody, preventing binding and entry of pathogen to the host cell receptor [significance].
 - b. Opsonisation* where Fc portion of antibody binds to Fc receptors on phagocytes to promote phagocytosis of the pathogen [significance].
 - c. Agglutination* where each antibody binds to 2 pathogens simultaneously, causing clumping/aggregation of pathogens and promoting phagocytosis of the pathogen [significance].
 - d. Hinge region* confers flexibility to the antibody to allow it to bind to antigens variable distances apart.

C. MHC proteins

1. Binds to peptides derived from pathogens to form peptide:MHC complexes*.
2. Peptide:MHC complexes* on APC/naïve B cell allow for recognition by naïve T cell receptor/helper T cell receptor respectively.
3. This recognition allows for clonal expansion and differentiation of naïve T and B lymphocytes into helper/cytotoxic/memory T cells and plasma/memory B cells. [significance].

6. Complexes

Protein-RNA complexes

A. Spliceosomes

1. Spliceosomes recognize and bind to specific splice sites and excise introns and join exons to form a mature RNA* for translation [significance].
2. Alternative splicing* where introns are excised and some exons are joined by spliceosomes in different combinations to produce a single pre-mRNA which can produce different mature mRNA*/proteins, resulting in more than one type of polypeptide [significance].

Other rejected proteins: telomerase, haemoglobin, collagen, proteasome.

QWC: 3 of the 5 categories must be covered.

(b) Compare gene mutation with chromosomal aberration.

[10]

Differences

		Gene mutation	Chromosomal aberration
D1	Nucleotide sequences	<u>Alteration/changes in a sequence of nucleotides on DNA in a particular gene;</u>	<u>Alteration/changes in structure/ large segment of chromosome;</u> Reject gene/genetic segment
D2	Chromosome number	<u>No change in chromosome number;</u>	May result in <u>change in chromosome number e.g. aneuploidy/polyploidy;</u>
D3	Number of gene changes	<u>Each mutation involves changes in one gene/gene locus;</u>	<u>Each mutation may involve changes in more than one gene/ gene loci;</u>
D4	Caused by	<u>Caused by random error during DNA replication or damage to DNA due to radiation and mutagens;</u>	<u>Caused by in crossing over during prophase I / meiosis I;</u> <u>Or</u> <u>Non disjunction of chromosomes/ homologous chromosomes failed to separate during anaphase I/meiosis I.</u> <u>Accept if candidate describe non disjunction of chromatids in anaphase II</u> <u>Or</u> <u>Segment of chromosome break and rejoin to another chromosome.</u>
D5	Time point where it occurs	Anytime Accept S phase of interphase/DNA replication	During mitosis/meiosis
D6	Effect	<u>Can result in silent mutation/no effect on phenotype</u>	<u>Usually will cause disease/change in phenotype</u> <i><u>Pls note that this is not always true but this is an acceptable answer considering the scope of the H2 syllabus coverage.</u></i>
D7	Effects on alleles	<u>May generate new types of alleles</u>	<u>Does not generate new types of alleles</u> <u>Or</u> <u>Generate new combination of alleles on chromosomes</u>
D8	Detected with karyotype	<u>Cannot be observed on karyotype</u>	<u>Can be observed on a karyotype</u>
<i>1 mark for each contrasting point.</i>			

Reject answers :

- that gives a list of specific types of gene mutation and chromosomal mutations (see comments below)
- that describes in detail the effect/mechanism of specific types of gene and chromosomal mutation and how it occurs (see comments below)
- that gives a list of diseases caused (see comments below)
- that state that gene mutation affect coding regions and chromosome mutation affect non coding region. (overgeneralisation)
- that states that frequency of gene mutation is high vs frequency of chromosomal mutation is low. (overgeneralisation, evidence of this?)

Similarities

S1	Both may involve <u>changes/alterations</u> in <u>DNA nucleotide sequence/genotype</u> ; Reject genetic sequence;
S2	Both may <u>result in abnormalities/disease/cancer</u> ; Reject negative effects.
S3	Both may result in <u>changes</u> in <u>proteins produced/phenotype</u> ;
S4	Both may give rise to new allele/ <u>variation in a population</u> ;
S5	Both may affect <u>germ line and somatic cell</u> ;
S6	Both may be <u>inherited by offspring</u> ;
S7	Both may be caused by <u>radiation/mutagen</u> ;
S8	Both may arise spontaneous/randomly

QWC - must have at least 2 from similarities and 2 from differences.

Max 4 marks if only similarities or only differences

[Total: 25]

- 5 (a) Discuss how pathogens (apart from the Severe Acute Respiratory Syndrome Coronavirus 2), lifestyle and genetics can cause diseases in humans. [15]

D1	Define disease: a condition that results in <u>ill-health and abnormal functioning of human body</u> ;
Pathogens (Max 8m for section on Pathogens)	
P1	Pathogens cause disease by <u>damaging cells</u> and <u>producing toxins</u> ;
P2	<u>Bacteria*</u> , <u>viruses*</u> and fungi are pathogens;
<u>Tuberculosis</u>	
PM1	<u>Mycobacterium tuberculosis</u> causes pulmonary <u>tuberculosis</u> ;
PM2	transmitted in <u>fine aerosol droplets</u> when an infected person with active TB <u>sneezes/coughs</u> & an <u>uninfected person inhales the droplets</u> ;
PM3	Once bacteria is inside lungs, alveolar <u>macrophages phagocytose the bacteria</u> ;
PM4	In the phagosome, <u>bacteria inhibits fusion of phagosome with lysosome and no phagolysosome formed</u> , no lysosomal enzymes kill the bacteria;
PM5	Bacteria survives, continues to <u>multiply inside macrophage</u> that phagocytosed them;
PM6	Macrophages secrete cytokines and <u>recruit more macrophages</u> to infected alveolus and <u>tubercle/ball-like clustering of infected cells surrounded by immune cells like macrophages</u> is formed;
PM7	At the centre of the tubercle, <u>cell death by necrosis</u> occurs and this involves rupturing of macrophage cell membrane & <u>releasing</u> of cell contents (i.e. <u>bacteria</u>) into the <u>tubercle cavity</u> ;
PM8	<u>Disease may be arrested at this stage</u> & remain <u>latent for years</u> as persons who have TB infection, but not TB disease, do not spread disease to others;
PM9	<u>TB disease occurs</u> when the <u>tubercle ruptures</u> , the <u>bacteria spills into a bronchiole</u> & <u>spreads throughout the lungs</u> <u>leading to productive cough</u> that facilitates aerosol <u>spread of bacteria</u> ;
PM10	<u>Rupturing of tubercles</u> also leads to formation of <u>cavities in lungs</u> and hence <u>lungs progressively destroyed</u> ;
<u>Flu</u>	
PI1	<u>Influenza virus</u> can cause the <u>flu illness</u> ;
PI2	<u>Transmission is by droplets</u> of moisture from lungs of <u>infected person and uninfected person inhales the droplets</u> ;
PI3	When influenza <u>haemagglutinin</u> will bind to <u>sialic acid receptors</u> on <u>epithelial cells of respiratory tract</u> ;
PI4	

PI5	Ref to replication of new influenza virus in epithelial cell. Accept if candidate gave detailed description of replicative cycle;
PI6	New influenza virus then <u>buds off</u> , infected <u>host</u> epithelial cells eventually <u>lyse</u> . The build up of dead epithelial cells results in inflammation and <u>symptoms</u> of influenza appear <u>runny nose & scratchy throat</u> ;
	The epithelial layer weakens and the individual is more <u>susceptible to bacterial infections like pneumonia</u> ;
PH1	<u>Acquired Immunodeficiency Syndrome</u> <u>Human Immunodeficiency virus (HIV)</u> can cause the <u>Acquired Immunodeficiency Syndrome</u> ;
PH2	<u>Transmission by sexual contact with infected person/sharing of needles used by infected person/infected mother to foetus</u> ;
PH3	When <u>HIV GP120</u> binds to <u>CD4 receptor on a T helper cell</u> ;
PH4	Ref to replication of new HIV virus in epithelial cell. Accept if candidate gave detailed description of replicative cycle;
PH5	Infected host <u>T helper cells</u> eventually <u>lyse</u> . With <u>fewer T helper cells</u> , the <u>immune system is depressed</u> & individuals are <u>more susceptible to opportunistic infections</u> . When infections become unmanageable leading to AIDS and finally death;
AVP	Other pathogen and disease if candidate can give relevant and correct details;
Lifestyle (Max 4)	
L1	<u>Activities that exposes individual to sun/UV exposure</u> eg. prolong sunbathing without protection;
L2	Cigarettes <u>smoking</u> results in <u>exposure to carcinogens</u> found in smoke;
L3	Resulting in <u>DNA damage/causes mutation</u> , <u>increasing risk of cancer</u> ;
L4	<u>Stress/obesity/sleep deprivation</u> can <u>compromise the immune system</u> and lead to increased <u>susceptibility to infections</u> ;
L5	<u>Working in an environment with high pathogen levels</u> eg. hospitals <u>without adequate protection</u> eg gloves, mask, will also make one more <u>susceptible to infections</u> ;
L6	<u>Diet high in cholesterol</u> increases risk of coronary heart disease/atherosclerosis;
AVP	Other lifestyle and disease if candidate can give relevant and correct details;
Genetics (Max 8)	
G1	Mutations and the inheritance of mutations in the genes can cause diseases; Examples of inherited disease are:
G2	<u>Sickle cell anaemia</u>

G3	<u>Sickle cell anaemia</u> in which the condition is <u>autosomal recessive</u> cause by a <u>substitution mutation / change from GAG to GTG</u> on beta globin gene;
G4	Changes codon on mRNA from GAG to GUG;
G5	6 th amino acid, <u>hydrophilic glutamic acid replaced with hydrophobic valine</u> ;
G6	Ref to detrimental effect of sickle cell anaemia;
G7	<u>Cancer</u> Cancer can occur when there is <u>an accumulation of mutations in a single cell</u> ;
G8	<u>Proto-oncogenes</u> <u>Gain-in-function mutations in proto-oncogenes (eg Ras gene)</u> ;
G9	Ref to relevant effects of mutation in proto-oncogenes;
G10	<u>Tumour suppressor gene</u> <u>Loss-of-function mutations of tumour suppressor genes (eg p53 gene)</u> ;
G11	Ref to relevant effects of mutation in tumour suppressor genes;
G12	Description of effect on cell cycle/uncontrolled cell division R uncontrolled cell growth;
G13	<u>Other genes involves in cancer</u> <u>Gain in function mutation of other genes involved in cell cycle/E.g</u> <u>Telomerase/cyclin/cyclin dependent kinases</u>
G14	Ref to relevant effects of mutation;
G15	Description of effect on cell cycle/uncontrolled cell division R uncontrolled cell growth;
G16	<u>Down's syndrome</u> <u>Ref to non-disjunction of chromosome during anaphase I/meiosis I or Ref to</u> <u>non-disjunction of chromatid during anaphase II/meiosis II</u> ;
G17	2 copies chromosome 21 in gametes;
G18	Fertilisation with another gamete, result in <u>zygote with 3 chromosome 21/trisomy 21</u> resulting in <u>Down's syndrome</u>
AVP	Other genetic disease if candidates can give relevant and correct details. E.g colour blindness, haemophilia, BRCA.
QWC: 1m for correct answers in each other 3 categories, pathogen, lifestyle and genetics.	

(b) Describe and explain the difference between continuous and discontinuous genetic variation. [10]

Differences:

Point of comparison	Continuous variation	Explanation	Discontinuous variation	Explanation
1. No. of genes	Continuous variation usually is the combined effect of <u>many genes/polygenes</u> controlling the characteristic		Discontinuous variation involves <u>one or a few genes</u> controlling the characteristic;	
2. Alleles	Many alleles		Few alleles;	
3. Additive effect	There is an <u>additive effect</u> * of two or more (usually many) genes on a single phenotypic character	a. Many genes/ alleles b. Different alleles have a small effect on phenotype	There is no additive effect;	a. Few genes/allele s b. Alleles have a large effect on phenotype
4. Effect of alleles	<u>Different alleles</u> have a <u>small effect on phenotype</u>	a. Many alleles	<u>Different alleles</u> have a <u>large effect on phenotype</u> ;	a. Few alleles
5. Type of curve	Result in a <u>bell-shape / normal distribution curve</u>	a. Additive effect b. Range of phenotypes c. Quantitative d. Intermediate phenotypes more common than extreme phenotypes	Result in a <u>bar graph</u> ;	a. No additive effect b. Discrete phenotypes c. Qualitative d. No intermediates
6. Nature of character	Quantitative		Qualitative;	
7. Phenotypes	Consists of unbroken <u>range of phenotypes/no discrete phenotypes</u>	a. Phenotypes affected by environmental factors b. Additive effect	Consists of <u>discrete phenotypes</u> ;	a. Phenotypes not affected by environmental factors b. No additive effect

		c. Many genes/alleles		c. Few genes/alleles
8. Intermediate phenotypes	<u>Intermediate phenotypes</u> are more common than extreme phenotypes;	a. Many alleles/genes b. Phenotypes affected by environmental factors	There are <u>no intermediates</u> ;	a. Few alleles/genes b. Phenotypes not affected by environmental factors
9. Effect of environment	Phenotypes are <u>affected by environmental factors</u> ;		<u>Environment has little/no effect</u> on phenotype;	
10. Example	Skin colour/ height/ weight;		Blood group/flower colour/leaf shape;	

Each point – 1m for each contrasting point

QWC: 1m – all comparisons must be point for point

[Total: 25]

End of Paper